

FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – III

JEE (Main)-2022

TEST DATE: 01-06-2022

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A** and **Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

Physics

PART – A

SECTION – A (One Options Correct Type)

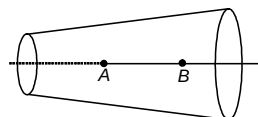
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A particle moving in a straight line is acted by a force, which works at a constant power and changes its velocity from u to v in passing over a distance x . The time taken will be:

- (A) $x \left(\frac{v-u}{v^2+u^2} \right)$
 (B) $x \left(\frac{v+u}{v^2+u^2} \right)$
 (C) $\frac{3}{2} x \left(\frac{v^2-u^2}{v^3-u^3} \right)$
 (D) $x \left(\frac{v}{u} \right)$

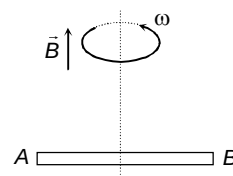
2. A resistance of frustum shape is shown in figure. If a current i passes through the resistance, the electric field at A and B are related as

- (A) $E_A > E_B$
 (B) $E_B > E_A$
 (C) $E_A = E_B$
 (D) There is no relation

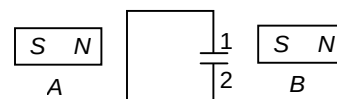


3. A conducting rod of length $2l$ is rotating with constant angular speed ω about its perpendicular bisector. A uniform magnetic field $\vec{B}$ exists parallel to the axis of rotation. The emf induced between two ends of the rod is

- (A) $B\omega l^2$
 (B) $\frac{1}{2} B\omega l^2$
 (C) $\frac{1}{8} B\omega l^2$
 (D) Zero

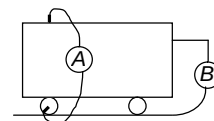


4. Two bar magnet moving with same speed in the given figure. Choose correct statement.



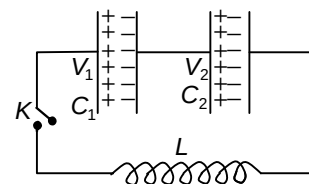
- (A) Plate (1) will be +ve relative to (2) if A moving B moving towards left.
 (B) Plate (1) will be +ve relative to (2) if A moving towards left and B moving towards right.
 (C) Charge on capacitor increase if A and B are at rest.
 (D) None of these

5. A flat, rail road car with a wooden base moves left with some velocity. A millivoltmeter A is connected across the axle of the car and is in the car. Another voltmeter B is connected across the rails of the car. Then



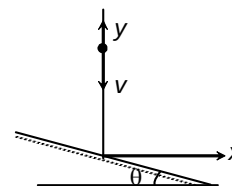
- (A) Reading of A and $B = \text{zero}$
 (B) Reading of $A = \text{zero}$ and B is non zero.
 (C) Reading of $B = \text{zero}$ and A is non zero.
 (D) Reading of A and B are non zero.

6. Two capacitors of capacitance C_1 and C_2 are charged to a potential difference of V_1 and V_2 respectively and are connected to an inductor of inductance L as shown in the figure. Initially key k is open. Now key k is closed and current in the circuit starts increasing. When current in the circuit is maximum



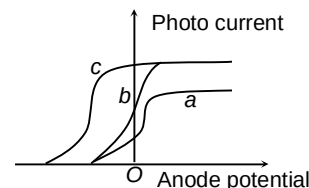
- (A) charge on both the capacitors is same
 (B) induced emf in the inductor is zero
 (C) potential difference across both the capacitors is half of the induced emf
 (D) electrostatic potential energy stored in both the capacitors is same

7. An object is moving towards a mirror with a velocity v as shown in figure. If the collision between the mirror and the object is perfectly elastic, then the velocity of the image after collision with mirror in vector form is



- (A) $-v\hat{j}$
 (B) $-v \cos 2\theta \hat{j} + v \sin 2\theta \hat{i}$
 (C) $-v\hat{i}$
 (D) $-v \cos 2\theta \hat{j} - v \sin 2\theta \hat{i}$

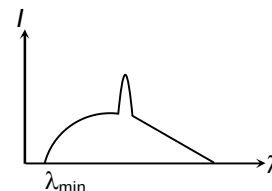
8. The figure shows the variation of photo current with anode potential for a photo-sensitive surface for three different radiations. Let I_a , I_b and I_c be the intensities and f_a , f_b and f_c be the frequencies for the curves a , b and c respectively;



- (A) $f_a = f_b$ and $I_a \neq I_b$
 (B) $f_a = f_c$ and $I_a = I_c$
 (C) $f_a = f_b$ and $I_a = I_b$
 (D) $f_b = f_c$ and $I_b = I_c$

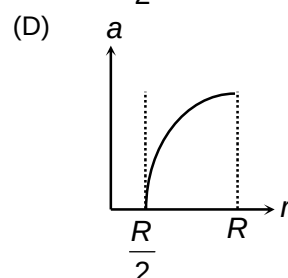
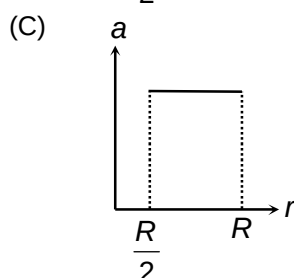
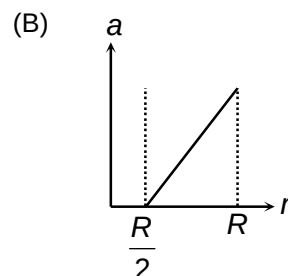
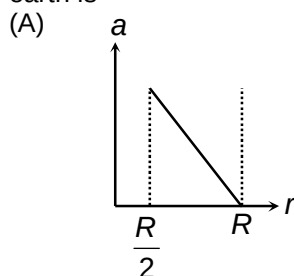
9. An X-ray tube has three main controls.
 (i) the target material (its atomic number Z)
 (ii) the filament current (I_f) and
 (iii) the accelerating voltage (V)

Figure shows a typical intensity distribution against wavelength. Which of the following is incorrect?



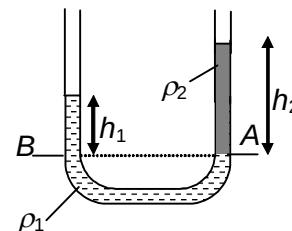
- (A) The limit $\lambda_{\min}$ is proportional to V^{-1}
 (B) The sharp peak shifts to the right as Z is increased
 (C) The penetrating power of X ray increases if V is increased
 (D) The intensity everywhere increases if filament current I_f is increased

10. A frictionless tunnel is dug along a chord of the earth at a perpendicular distance $R/2$ from the centre of earth (where R is radius of earth). An object is released from one end of the tunnel. The correct graph, showing the variation of acceleration of particle with its distance r from centre of earth is



11. Two immiscible liquids are poured in a U-tube having densities ρ_1 and ρ_2 . The ratio of height of the liquids above their interface (h_1/h_2) is

- (A) directly proportional to their densities
 (B) inversely proportional to their densities
 (C) directly proportional to square of their densities
 (D) equal

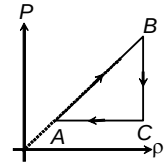


12. Two particles are performing simple harmonic motion with same amplitude and same frequency. When they are at same distance from mean position on opposite sides, their speeds are one fourth of maximum speed. Also at these positions their direction of velocities are opposite, the phase difference between the two simple harmonic motions is

- (A) $\frac{\pi}{2}$
 (B) π
 (C) $\frac{3\pi}{2}$
 (D) $\frac{5\pi}{4}$



13. Pressure versus density of an ideal gas is shown in the graph. Then the correct temperature versus density of gas graph is



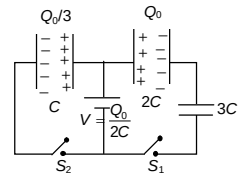
- (A)
- (B)
- (C)
- (D)

14. P-V diagram of a diatomic gas is straight line parallel to P-axis. The molar heat capacity of the gas in the process will be

- (A) $4R$
 (B) $2.5 R$
 (C) $3R$
 (D) $4R/3$

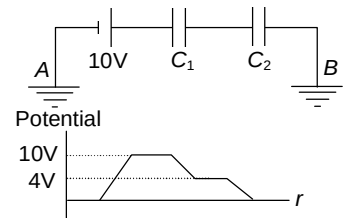
15. In the given circuit, the initial charges on the capacitors are shown in the figure. The charge flown through the switches S_1 and S_2 respectively after closing the switches are

- (A) zero, $\frac{Q_0}{6}$
 (B) $\frac{Q_0}{5}$, $\frac{Q_0}{2}$
 (C) zero, $\frac{Q_0}{2}$
 (D) $\frac{3}{5}Q_0$, $\frac{Q_0}{6}$



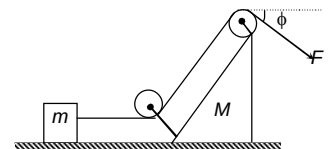
16. Figures shows two capacitors C_1 and C_2 connected with 10 V battery and terminal A and B are earthed. The graph shows the variation of potential as one moves from left to right. Then the ratio of C_1/C_2 is

- (A) $5/2$
 (B) $2/3$
 (C) $2/5$
 (D) $4/3$



17. In the given arrangement pulleys and string are massless and frictionless and all surfaces are smooth. Find the magnitude of acceleration of wedge of mass M .

- (A) $\frac{F \cos \phi}{M}$



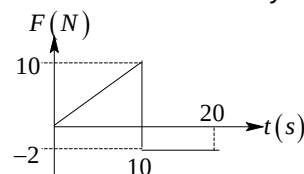
- (B) $\frac{F(1 - \cos \phi)}{M}$
- (C) $\frac{F}{m + M}$
- (D) $\frac{F \cos \phi}{m + M}$

18. Mark the incorrect statement among the following:

- (i) a particle can have zero velocity and non-zero acceleration
 (ii) a particle can have zero displacement and non-zero velocity
 (iii) a particle can have zero acceleration and non-zero velocity
 (iv) a particle can have zero displacement and non-zero average velocity

- (A) (i)
 (B) (ii)
 (C) (iii)
 (D) (iv)

19. The force time graph for the motion of a body is shown in the following figure. The change in momentum of the body between 0 and 20 s is:



- (A) 20 kg ms^{-1}
 (B) 25 kg ms^{-1}
 (C) 30 kg ms^{-1}
 (D) 35 kg ms^{-1}

20. Under the action of a force $\vec{F} = xy^2\hat{i} + yx^2\hat{j}$, a particle is moving along a parabolic path given by $y = x^2$. The work done by this force in moving the particle from $(0, 0)$ to (a, a^2) is:

- (A) $\frac{a^6}{2}$
 (B) $\frac{a^5}{2}(a^2 + 2)$
 (C) $\frac{a^4}{2}$
 (D) $a^2 + a^3$

SECTION – B
(Numerical Answer Type)

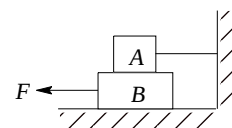
This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXX.XX).

21. A particle is moving along x-axis whose acceleration is given by $a = 3x - 4$, where x is the position of the particle. At $t = 0$, the particle is at rest at $x = \frac{4}{3}m$. The distance travelled by the particle in 5 s is (in m) ____

22. A birds flies for 4 s with a velocity of $|t - 2| \text{ ms}^{-1}$ in a straight line, where t is in second. The distance travelled by it is (in m) ____

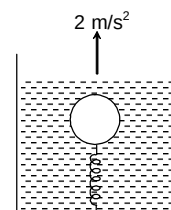
23. A fire hose squirts 12 kg s^{-1} of water against a flat plate (normally). The velocity of stream is 10 ms^{-1} . If the water flows parallel to the plate after striking it, the average force on the plate is (in N) ____

24. Block A weighs 4 N and block B weighs 8 N. The coefficient of kinetic friction is 0.25 for all surfaces. The force F if B slides at constant speed with A at rest on B and moves with it, is (in N) ____.

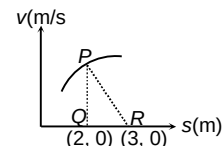


25. A uniform rope of length 12 m and mass 6 kg hangs vertically from a rigid support. A block of mass 2 kg is attached to the free end of the rope. A transverse pulse of wavelength 0.06 m is produced at the lower end of the rope. The wavelength of the pulse when it reaches the top of the rope is (in m) ____

26. A ball of mass 10 kg and density 1 gm/cm^3 is attached to the base of a container having a liquid of density 1.1 gm/cm^3 , with the help of a spring as shown in the figure. The container is going up with an acceleration 2 m/s^2 . If the spring constant of the spring is 200 N/m, the elongation in the spring is (in cm)



27. Velocity versus displacement curve of a particle moving in straight line is shown in the figure. From a point P , a line is drawn perpendicular to displacement axis and line PR is drawn normal to the curve at P . The instantaneous acceleration of the particle at point P is (in m/sec^2)



28. A brass scale is graduated at 10°C . What is the true length of a zinc rod (in S.I. unit) which measures 60.00 cm on this scale at 30°C . Coefficient of linear expansion of brass $= 18 \times 10^{-6} \text{ K}^{-1}$.

29. A very small sphere of mass 80 g having a charge q is held at a height 9 m vertically above the centre of a fixed non conducting sphere of radius 1 m, carrying an equal charge q . When released it falls until it is repelled just before it comes in contact with the sphere. Calculate the charge q . (in μC) $[g = 9.8 \text{ m/s}^2]$

30. In a two slit experiment with monochromatic light, fringes are obtained on a screen placed at some distance from the plane of slits. If the screen is moved by $5 \times 10^{-2} \text{ m}$ towards the slits, the change in fringe width is $3 \times 10^{-5} \text{ cm}$. If the distance between the slits is 10^{-3} m , calculate the wavelength of the light used (in $\text{\AA}$) ____.

Chemistry

PART – B

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. The volume of '10 Volume' of H_2O_2 required to liberate 500 mL O_2 at NTP is:
 (A) 50 mL
 (B) 25 mL
 (C) 100 mL
 (D) 125 mL
32. On the addition of mineral acid to an aqueous solution of borax, the compound formed is:
 (A) Orthoboric acid
 (B) Boron hydride
 (C) Metaboric acid
 (D) Pyroboric acid
33. On heating graphite with conc. HNO_3 repeatedly, a yellow mass is obtained which is called:
 (A) Graphitic oxide
 (B) Graphitic peroxide
 (C) Benzene hexacarboxylic acid
 (D) Graphitic nitrate
34. $\text{BaC}_2 + \text{N}_2 \xrightarrow{\Delta} (1)$
 $\text{CaC}_2 + \text{N}_2 \xrightarrow{\Delta} (2)$
 (1) and (2) are:
 (A) BaCN_2 , CaCN_2
 (B) $\text{Ba}(\text{CN})_2$, $\text{Ca}(\text{CN})_2$
 (C) $\text{Ba}(\text{CN})_2$, CaCN_2
 (D) None is correct
35. Consider the reaction

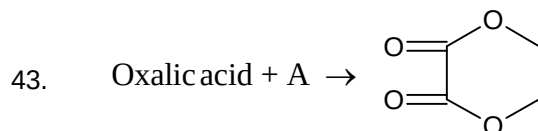
$$\left[\text{H}_2\text{CH}_2\text{CH}_3\text{C}-\overset{\overset{\text{CH}_3}{|}}{\underset{\underset{\text{CH}_3}{|}}{\text{N}^+}}-\text{CH}_2\text{CH}_3 \right] \text{OH}^- \xrightarrow{\text{heat}}$$

 Which of the following is formed in major amount?
 (A) $\text{CH}_2 = \text{CH}_2$
 (B) $\text{CH}_3\text{CH} = \text{CH}_2$
 (C) Both (A) and (B) in equal amount
 (D) None, as no reaction takes place.

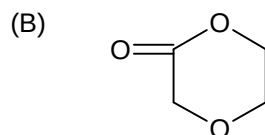
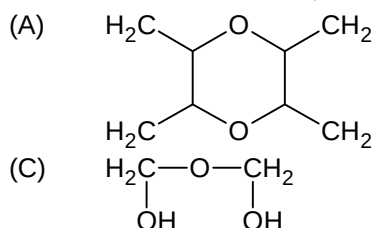
36. Consider the following statements about benzene:
(I) Heats of hydrogenation of benzene and 1, 3, 5-cyclohexatriene are identical
(II) Benzene is much more stable than expected for 1, 3, 5-cyclohexatriene
(III) All carbon-carbon bonds (single and double) have equal length
Select correct statements:
(A) I, II
(B) II, III
(C) I, III
(D) I, II, III
37. Lassaigne's test for the detection of nitrogen fails in:
(A) $\text{NH}_2\text{CONHNH}_2 \cdot \text{HCl}$
(B) $\text{NH}_2\text{NH}_2 \cdot \text{HCl}$
(C) NH_2CONH_2
(D) $\text{C}_6\text{H}_5\text{NHNH}_2 \cdot \text{HCl}$
38. Among the following ions, which one has the highest paramagnetism?
(A) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
(B) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
(C) $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$
(D) $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$
39. In blast furnace, the hearth is lined with:
(A) Dolomite refractories
(B) Alumina refractories
(C) Chromite refractories
(D) Carbon refractories
40. Ca, Ba and Sr ions are precipitated in fifth group as their:
(A) Oxides
(B) Sulphates
(C) Carbonates
(D) Sulphides
41. Of the following statements:
(P) $\text{C}_6\text{H}_5\text{N}=\text{CH}-\text{C}_6\text{H}_5$ is a Schiff's base
(Q) A dye is obtained by the reaction of aniline and $\text{C}_6\text{H}_5\text{N}=\text{NCl}$
(R) $\text{C}_6\text{H}_5\text{CH}_2\text{NH}_2$ on treatment with $[\text{NaNO}_2 + \text{HCl}]$ gives diazonium salt
(S) p-Toluidine on treatment with $[\text{HNO}_2 + \text{HCl}]$ at 2°C gives diazonium salt.
(A) Only (P) and (Q) are correct
(B) Only (P) and (R) are correct
(C) Only (R) and (S) are correct
(D) (P), (Q) and (S) are correct

42. When sucrose is heated with conc. HNO_3 the product is:

- (A) Sucrose nitrate
 (B) Oxalic acid
 (C) Formic acid
 (D) Citric acid



Hence A $\xrightarrow{\text{conc. H}_2\text{SO}_4}$ B, B is:



(D) None of these

44. A gas at a pressure of 5.0 atm is heated from 0° to 546°C and simultaneously compressed to one-third of its original volume. Hence final pressure is:

- (A) 10.0 atm
 (B) 30.0 atm
 (C) 45.0 atm
 (D) 5.0 atm

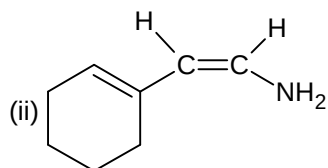
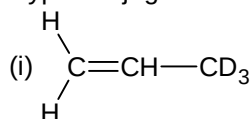
45. Match compounds given in (X) with their uses in (Y):

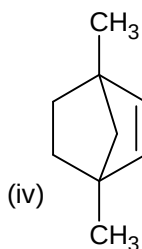
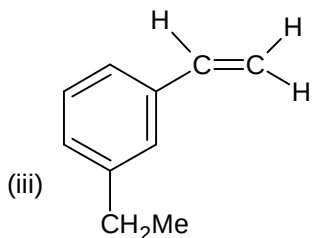
- | X | Y |
|------------------------------|---------------------------|
| (a) Na_2CO_3 | 1. Glass |
| (b) Na_2SO_4 | 2. Bleach |
| (c) NaOH | 3. SO_2 Absorber |
| (d) NaOCl | 4. Detergent |

Hence correct order is:

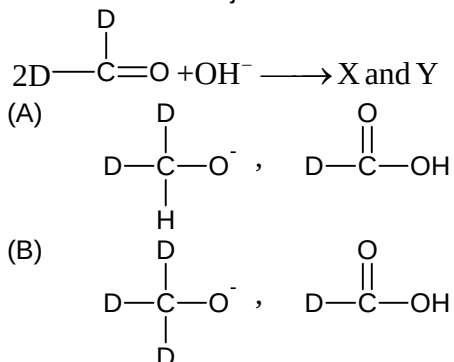
- | | a | b | c | d |
|-----|---|---|---|---|
| (A) | 1 | 4 | 3 | 2 |
| (B) | 1 | 3 | 4 | 2 |
| (C) | 2 | 4 | 1 | 3 |
| (D) | 3 | 2 | 4 | 1 |

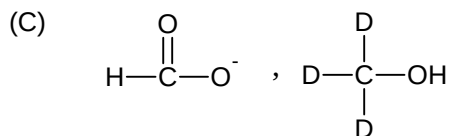
46. Hyperconjugation is possible in:





- (A) i and ii
(B) i, ii and iii
(C) only ii
(D) in all of these
47. Metals like Pt and Pd can take up large volume of hydrogen under special conditions. Hydrogen thus retained by the metal is called:
(A) Absorbed hydrogen
(B) Nascent hydrogen
(C) Reactive hydrogen
(D) Occluded hydrogen
48. The complex $\text{Hg}[\text{Co}(\text{CNS})_4]$ is correctly named as:
(A) Mercury tetrathiocyanato cobaltate (II)
(B) Mercury cobalt tetrasulphocyano (II)
(C) Mercury tetrasulphocyanide cobaltate (II)
(D) Mercury sulphocyanato cobalt (II)
49. A monoprotic acid in a 0.1 M solution ionizes to 0.001%. Its ionisation constant is:
(A) 1.0×10^{-3}
(B) 1.0×10^{-6}
(C) 1.0×10^{-8}
(D) 1.0×10^{-11}
50. On the basis of major concentration X and Y will have given below structure:



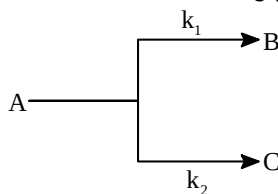


(D) None is correct

SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXX.XX).

51. Gold crystallizes in a face centred cubic lattice. If the length of the edge of the unit cell is 407 pm. The density of gold assuming it to be spherical is g/cm³. Atomic mass of gold = 197 amu. (Take $N_A = 6 \times 10^{23}$)
52. The degree of dissociation of $\text{Ca}(\text{NO}_3)_2$ in a dilute aqueous solution containing 7g of the salt per 100g of water at 100°C is 70%. If the vapour pressure of water at 100°C is 760mm. The vapour pressure of the solution is mm after rounding off to nearest integer.
53. Through an aqueous solution of an unknown salt of metal M ($M = 200$ g/mol) a current of 1.93A is passed for 50 min. If 4g of metal is produced at cathode. The charge on metal ion in solution is.....
54. A sample U^{238} (half life= 4.5×10^9 years) ore is found to contain 23.8g of U^{238} and 61.8g of Pb^{206} . The age of the ore is $\times 10^9$ years.
55. Total sodium ions which are present in one formula unit of sodium ethane-1, 2-diaminetetraacetatochromate (II) and sodium hexanitrito cobaltate (III) is_____.
56. The heat of reaction for an endothermic reaction in equilibrium is 1200 cal more at constant volume than at constant pressure at 300K. Calculate the ratio of equilibrium constants K_P and K_C . ($R = 2$ cal or 0.0821 in terms of L atm) if it is $a \times 10^{-3}$ then the value of a is_____.
57. The pH of 2.0×10^{-4} M H_3X solution assuming first dissociation to be 100%, second to be 50% and third to be negligible is_____.
58. Consider the following path followed by A:



Calculate $[\text{C}]$ in millimoles per L after 3h of the reaction if $[\text{A}]_0 = 0.25 \text{ mol L}^{-1}$. Also given that $k_1 = 2.0 \times 10^{-5} \text{ s}^{-1}$ and $k_2 = 4.0 \times 10^{-5} \text{ s}^{-1}$. The value of antilog (2.814) is 19.1.

59. The total number of enantiomer pairs formed by the isomer of $[\text{M}(\text{AB})_3\text{b}]$, $[\text{M}(\text{AB})_2\text{b}_2]$, $[\text{M}(\text{AB})_2\text{bc}]$
60. How many are colourless among following ClF , ClF_3 , BrCl , ClF_5 (g), IF_7 (g), BrF_5 (l), ICl (s), IF_5 (l)

Mathematics**PART – C****SECTION – A**
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. If $a, b (a > b > 0)$ be 2 numbers such that the ratio of their H.M. and G.M. is $(4r^2 - 1) : (4r^2 + 1)$ and r is so large that square and higher powers of $\frac{1}{r}$ may be neglected then $\sqrt{a} : b$ is equal to
- (A) $\frac{r-1}{r}$
 (B) $\frac{r+1}{r}$
 (C) $1 - \frac{1}{2r}$
 (D) None of these
62. If ω and ω^2 satisfy the equation $\frac{1}{a+x} + \frac{1}{b+x} + \frac{1}{c+x} + \frac{1}{d+x} = \frac{2}{x}$ then value of $abcd$ is
- (A) 3
 (B) 0
 (C) 2
 (D) -1
63. $\int \frac{dx}{\sin x (2 + \cos x - 2 \sin x)}$ is equal to
- (A) $\ln \left| \tan \frac{x}{2} \right| + \ln \left| \tan \frac{x}{2} + 1 \right| + c$
 (B) $\ln \left| \tan \frac{x}{2} \right| + \frac{5}{3} \ln \left| \tan \frac{x}{2} - 3 \right| - \ln \left| \tan \frac{x}{2} - 1 \right| + c$
 (C) $\frac{1}{3} \ln \left| \tan \frac{x}{2} \right| + \frac{5}{3} \ln \left| \tan \frac{x}{2} - 3 \right| - \ln \left| \tan \frac{x}{2} - 1 \right| + c$
 (D) None of these
64. The mean deviation from mean of the observation $a, a+d, a+2d, \dots, a+2nd$ is
- (A) $\frac{n(n+1)}{3} d^2$
 (B) $\frac{n(n+1)}{2} d^2$
 (C) $a + \frac{n(n+1)}{2} d^2$
 (D) None of these

65. If $\vec{a} = \vec{x} - \vec{y}$; $\vec{a} + \vec{b} = 2\vec{x}$, $|\vec{x}| = |\vec{y}| = 2$ and $|\vec{a} \times \vec{b}| = k\{k^{k^k} - (\vec{x} \cdot \vec{y})^2\}^{1/k}$ then $k =$
 (A) 1
 (B) $1/2$
 (C) 2
 (D) None of these
66. Find the number of integral values of h if one of the lines represented by $3x^2 \sin \theta + 2hxy + 4y^2 \cos \theta = 0$ bisects the angle between coordinate axes
 $\left[\theta \neq (2n+1)\frac{\pi}{2}, \text{ where } n \in \mathbb{Z} \right]$
 (A) 5
 (B) 2
 (C) 4
 (D) None of these
67. $\lim_{n \rightarrow \infty} \frac{[1] + [2] + [3] + \dots + [\sqrt{n^2 + n + 1}]}{n^2} = ([.] \text{ denotes greatest integer function})$
 (A) $1/2$
 (B) 0
 (C) Does not exist
 (D) None of these
68. The value of ${}^{404}C_4 - {}^4C_1 {}^{303}C_4 + {}^4C_2 {}^{202}C_4 - {}^4C_3 {}^{101}C_4$ is equal to
 (A) $(401)^4$
 (B) $(101)^4$
 (C) 0
 (D) $(201)^4$
69. The position vector of point of intersection of the planes $\vec{r} \cdot (\hat{n}_2 \times \hat{n}_3) = 1$, $\vec{r} \cdot (\hat{n}_3 \times \hat{n}_1) = 2$ and $\vec{r} \cdot (\hat{n}_1 \times \hat{n}_2) = 3$ ($[\hat{n}_1, \hat{n}_2, \hat{n}_3] \neq 0$) is
 (A) $\frac{\hat{n}_1 + 2\hat{n}_2 + 3\hat{n}_3}{[\hat{n}_1, \hat{n}_2, \hat{n}_3]}$
 (B) $\frac{\hat{n}_1 + 2\hat{n}_2 + 3\hat{n}_3}{[\hat{n}_1, \hat{n}_2, \hat{n}_3]^2}$
 (C) $\frac{\hat{n}_2 \times \hat{n}_3 + 2\hat{n}_3 \times \hat{n}_1 + 3\hat{n}_1 \times \hat{n}_2}{[\hat{n}_1, \hat{n}_2, \hat{n}_3]}$
 (D) $\frac{\hat{n}_2 \times \hat{n}_3 + 2\hat{n}_3 \times \hat{n}_1 + 3\hat{n}_1 \times \hat{n}_2}{[\hat{n}_1, \hat{n}_2, \hat{n}_3]^2}$

70. Let H, I and O be respectively the orthocentre, incentre and circumcentre of the triangle ABC. If $\angle HAI = \alpha$ and $\angle IAO = \beta$, then
- (A) $2\alpha = \beta$
 (B) $\alpha = 2\beta$
 (C) $\alpha = 4\beta$
 (D) $\alpha = \beta$
71. Let $C_k = {}^nC_k$ for $0 \leq k \leq n$ and $A_k = \begin{bmatrix} C_{k-1} & 0 \\ 0 & C_k \end{bmatrix}$ for $k \geq 1$ and $A_1 A_2 + A_2 A_3 + A_3 A_4 + \dots + A_{n-1} A_n = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ then
- (A) $a = b$
 (B) $a = b^2$
 (C) $a + b = 0$
 (D) $a = 2b$
72. A given right circular cone has volume p and the largest right circular cylinder that can be inscribed in the cone has a volume q. Then $p : q$ is
- (A) 9 : 4
 (B) 8 : 3
 (C) 7 : 2
 (D) None of these
73. The area bounded by the curve $f(x) = \left| \tan x + \cot x \right| - \left| \tan x - \cot x \right|$ between the lines $x = 0$, $x = \frac{\pi}{2}$ and the x-axis, is
- (A) $\ln \sqrt{2}$
 (B) $2 \ln \sqrt{2}$
 (C) $\ln 4$
 (D) $\sqrt{2} \ln 2$
74. Three numbers a, b, c are chosen randomly from the set of natural numbers. The probability that $a^2 + b^2 + c^2$ is divisible by 7 is
- (A) $1/3$
 (B) $1/4$
 (C) $1/5$
 (D) $1/7$

75. The reciprocal of the distance between two points one on each of the lines $\frac{x-2}{3} = \frac{y-4}{2} = \frac{z-5}{5}$ and $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$
- (A) cannot be less than 9
 (B) having minimum value $5\sqrt{3}$
 (C) cannot be greater than $\sqrt{78}$
 (D) cannot be $2\sqrt{19}$
76. The ellipse $4x^2 + 9y^2 = 36$ and the hyperbola $a^2x^2 - y^2 = 4$ intersect at right angles then the equation of the circle through the point of intersection of two conic is
- (A) $x^2 + y^2 = 5$
 (B) $\sqrt{5}(x^2 + y^2) = 3x + 4y$
 (C) $\sqrt{5}(x^2 + y^2) - 3x + 4y = 0$
 (D) $x^2 + y^2 = 25$
77. In an acute angled triangle ABC the circle on altitude AD as diameter cuts AB at P and AC at Q then $PQ =$ (Here R is circumradius, r is in radius, Δ is area and s is semi-perimeter)
- (A) R
 (B) $\frac{r}{\Delta}$
 (C) $\frac{\Delta}{R}$
 (D) Δs
78. If $f(x) = \begin{cases} \int_0^x (1 + |1-t|) dt & ; x > 2 \\ 5x - 7 & ; x \leq 2 \end{cases}$ then
- (A) $f(x)$ is not continuous at $x = 2$
 (B) $f(x)$ is differentiable everywhere
 (C) RHL at $x = 2$ does not exist
 (D) $f(x)$ is continuous but not differentiable at $x = 2$
79. Which of the following statements is a tautology?
- (A) $(\sim q \wedge p) \wedge q$
 (B) $(\sim q \wedge p) \wedge (p \wedge \sim p)$
 (C) $(\sim q \wedge p) \vee (p \wedge \sim p)$
 (D) $(p \wedge q) \wedge (\sim (p \wedge q))$

80. If $I_1 = \int_0^1 \frac{x^{5/2} (1-x)^{7/2}}{12} dx$; $I_2 = \int_0^1 \frac{x^{5/2} (1-x)^{7/2}}{(3+x)^8} dx$ then
- (A) $I_2 = 144\sqrt{3} I_1$
 (B) $I_1 = 864\sqrt{3} I_2$
 (C) $I_1 = I_2$
 (D) $I_1 I_2 = 144\sqrt{3}$

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. On a normal standard die one of the 21 dots from any one of the six faces is removed at random with each dot equally likely to be chosen. The die is then rolled. If the probability that the top face has an odd number of dots is $\frac{p}{q}$ where p and q are in their lowest form, find $\frac{(p+q)}{4}$
82. A function f is defined on the complex number by $f(z) = (a+bi)z$, where 'a' and 'b' are positive numbers. This function has the property that the image of each point in the complex plane is equidistant from that point and the origin. Given that $|a+bi| = 8$ and that $b^2 = \frac{u}{v}$ where u and v are coprimes. Find the value of (u + v)-250.
83. If $\begin{bmatrix} 1 & 2 & a \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}^n = \begin{bmatrix} 1 & 18 & 2007 \\ 0 & 1 & 36 \\ 0 & 0 & 1 \end{bmatrix}$ then find the value of $\frac{(n+a)}{100}$.
84. If the polynomial $f(x) = 4x^4 - ax^3 + bx^2 - cx + 5$ where $a, b, c \in R$ has four positive real roots say r_1, r_2, r_3 and r_4 , such that $\frac{r_1}{2} + \frac{r_2}{4} + \frac{r_3}{5} + \frac{r_4}{8} = 1$. Find the value of (a-10).
85. All the three vertices of an equilateral triangle lie on the parabola $y = x^2$, and one of its sides has a slope of 2. The x-coordinates of the three vertices have a sum equal to $\frac{p}{q}$ where p and q are relatively prime positive integers. Find the value of (q-p)
86. There are 4 bowlers, 4 batsman and 1 all-rounder (who is bowler as well as batsman). Randomly a team of 4 players consisting of at least two bowlers and at least two batsman is formed. If the probability that all-rounder has been selected in team is $\frac{4}{p}$, then p is equal to:

87. Suppose $\int \frac{1-7\cos^2 x}{\sin^7 x \cos^2 x} dx = \frac{g(x)}{\sin^7 x} + C$, where C is an arbitrary constant of integration. Then the value of $g'(0) + g''\left(\frac{\pi}{4}\right)$.
88. Mr. A lists all the positive divisors of the number $N(2010)^2$ and selects two divisors from the list then the probability that exactly one of the selected divisors is perfect squares is:
89. If $\frac{(x-3)^{\frac{-|x|}{x}} \sqrt{(x-4)^2} (17-x)}{\sqrt{-x}(-x^2+x-1)(|x|-32)} < 0$ then no. of integers x satisfying the inequality is:
90. Let f be real function defined on $\mathbb{R}$ (the set of real numbers) such that $f'(x) = 100(x-1)(x-2)^2(x-3)^3 \dots (x-100)^{100}$, for all $x \in \mathbb{R}$. If g is a function defined on $\mathbb{R}$ such that $\int_a^x e^{f(t)} dt = \int_0^x g(x-t) dt + 2x + 3$, If some of the all the values of x for which $g(x)$ has a local extremum be λ then find $\frac{\lambda}{3}$